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Abstract

A tapered slot jet was studied experimentally in nonreacting and reacting tests using hot-wire anemometry, water-tunnel flow visualization, and Planar Laser Induced Fluorescence (PLIF). The tapered slot jet is a modified elliptic jet which has a conical contraction leading to its outlet. The added contraction changes the entire flow field. The jet spread in the major axis plane is larger than in the minor axis plane, which is the opposite behavior of an elliptic jet. Consequently, no axes switching, typical to an elliptic jet, is observed. The turbulence amplification in the jet core is higher than in circular and elliptic jets. The different behavior is attributed to the change in flow direction, inside the nozzle, from the conical section to the slot outlet. During this transition the flow acquires angular momentum thereby generating axial vorticity. The influence of the contraction angle and the outlet aspect ratio were investigated. The effect of the augmented turbulence on reactive flow was tested in a premixed flame. The combustion rate was augmented in both the core and edges of the flame relative to a circular burner.

Introduction

The combustion process requires mixing between fuel and oxidizer and the existence of the right conditions to initiate and sustain the reaction between them. Efficient combustion can be achieved by optimizing the mixing and the reaction process. When the fuel and oxidizer are initially separated, it is of special importance to get sufficient large-scale mixing. After the reactants are brought together to the same volume, enhanced mixing in the molecular level is needed to accelerate the reaction rate. Both types of mixing are governed by the flow dynamics. In reacting systems which contain mixing layers, large-scale mixing is governed by large-scale coherent structures. These structures dominate the mixing layer evolution in both low and high Reynolds number flows.[1-3] Molecular mixing is related to the fine-scale turbulence which is produced during the development of the mixing layer.

Much research has been done to develop different active and passive methods for augmenting the large- and small-scale mixing. Recent reviews,[2,4] have summarized these efforts. One of the passive methods investigated was the use of small aspect ratio elliptic nozzles.[5] The large-scale mixing of this jet was substantially increased, relative to a circular jet, due to self induction of the elliptic vortices. The small-scale mixing was also enhanced using this jet as a result of high azimuthal instability modes amplified in this noncircular configuration. It was also shown to be effective in high Reynolds number reacting flows.[6] A modified configuration of the elliptic nozzle was developed later to gain additional small-scale mixing enhancement in the circumference and core region of the jet. The modified nozzle consists of a conical contraction region which leads into the elliptic outlet and is called the tapered elliptic nozzle. The basic flow characteristics of this jet are described in Gutmark and Schadow (1987).[7] The jet's growth pattern was very different from the elliptic jet. The jet spread in the major axis side more than to the minor axis side, which is the opposite of the elliptic jet growth characteristics. Consequently, the axes switching, typical to an elliptic jet, was not observed in the tapered elliptic jet. The growth rate of the turbulence level in the core region as well as in the jet's circumference was higher than in the circular and elliptic jets. The increased turbulence production was attributed to the axial vorticity component generated by the nozzle's design.

In the present paper the effect of the nozzle design parameters on the jet's flow field are studied in nonreacting flows, and the combustion characteristics of a tapered elliptic flame are visualized using Planar Laser Induced Fluorescence.

Experimental Set-Up

Three different experimental methods were used in this work. Nonreacting free-jet tests were performed to study the shear-flow dynamics with hot-wire anemometry. Flow visualizations were done in water tunnel tests. The tapered slot jet was studied in a laboratory burner, as well, using Planar Laser Induced Fluorescence (PLIF) imaging for combustion dynamic visualization.

The free jet tests were done using a tapered nozzle which had a conical contraction leading to the outlet elongated slot. The outlet slot replaced an elliptic shape to simplify the design and manufacturing. The slot consisted of a rectangular middle section with half circles, whose diameter was equal to the slot's width, attached to each end of the rectangle (Fig. 1). Five different tapered nozzles were tested in the air jet tests and compared to a circular nozzle. Their geometrical parameters are summarized in Table 1. The air was supplied by a blow down facility. The jet exit velocity, U_o, was typically 50 m/sec, corresponding to a Reynolds number of $1.6 \cdot 10^5$.

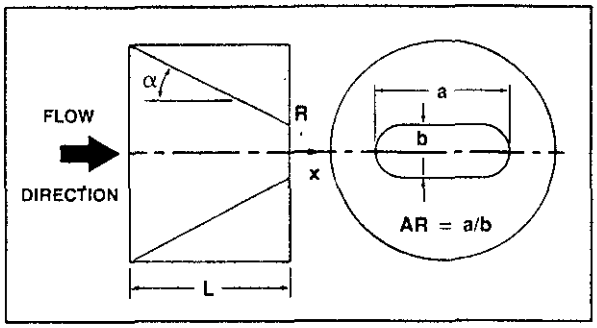

Fig. 1. Tapered Elliptic Nozzle Design.

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Table 1. Tapered Elliptic Nozzle Geometrical Configurations.

Nozzle Number		Geometrical Parameters			
		Aspect Ratio (a/b)	Length; L (cm)	Contraction Angle (α)	Equivalent Diameter; De (cm)
Air Tests	1	3:1	3.67	30°	3.95
	2	3:1	7.87	15°	3.95
	3	3:1	16.13	7.5°	3.95
	4	2:1	5.99	15°	4.76
	5	5:1	9.53	15°	3.28
Water Test	6	3:1	8.20	23°	4.80
Burner	7	3:1	2.54	22°	1.96

Hot-wire anemometers were used for probing the flow field; the measurements included mean velocity contours and the distribution of turbulent intensities. A computer controlled three-axes traverse mechanism was used. A typical scan of the flow field included 400 measurement points.

The hot-wire calibration, the data acquisition, and analysis were done on a VAX 750 minicomputer.

The flow visualization tests were done in a water tunnel using dye injection and air bubbles. The flow was illuminated with a slot lamp. Flow excitation was achieved using a butterfly valve in the inlet duct. The nozzle tested had the dimensions of Nozzle No. 6 in Table 1. The exit velocity was 0.5 m/s yielding Re = 20,000.

The flow characteristics of the propane/air tapered burner were visualized by PLIF imaging of in situ OH radicals. In this technique, which is schematically drawn in Fig. 2 and described in detail in Reference 8, an eximer laser beam is expanded into a thin sheet and excites OH molecules in the premixed flame of the burner. The fluorescence is imaged with an intensified diode array camera. The burner can be acoustically excited with varying phase shifts with respect to the laser. Changing the phase angle can be used as a means to stroboscopically map out the time dependence of the flame structure. The burner (Nozzle No. 7) had an aspect ratio of 3:1 and contraction angle of 22°. The air velocity was 5 m/sec corresponding to a Reynolds number of 6500.

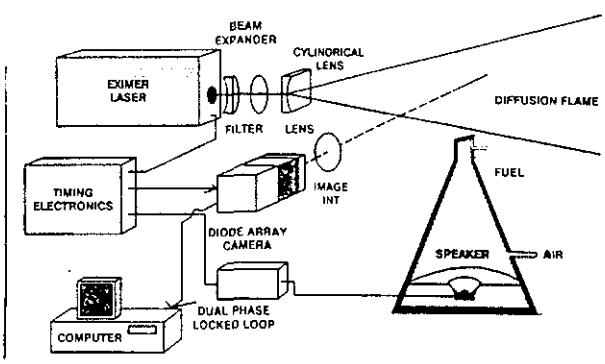

Fig. 2. Experimental Set-Up for PLIF OH-Imaging of a Burner.

Results and Discussion

Flow Visualization

Water tunnel tests were conducted to clarify the flow characteristics of the tapered slot jet both inside the nozzle and in the jet flow. Previous air tests results[7] showed that the tapered slot jet has a higher turbulence intensity in the jet center relative to a circular jet. The flow characteristics varied considerably around the jet's circumference. These features were attributed to the effect of the tapered nozzle on the outgoing flow. The flow streamlines inside the nozzle change from the conical contraction section to the slot outlet. During this transition the flow acquires a vorticity component in the axial direction. On the intersection region between the conical contraction and the slot, axial vortices are formed. This process is visualized inside the nozzle in Fig. 3. Two pairs of intense axial vortices are generated in the jet. One such pair is shown in Fig. 4 which was visualized by illuminating the jet along its longer dimension, off its axis. The axial vortices deform the outgoing circumferential vortices (Fig. 5), which then obtain complex azimuthal structure. The jet evolves into secondary jets located at the two sides of the major axis.

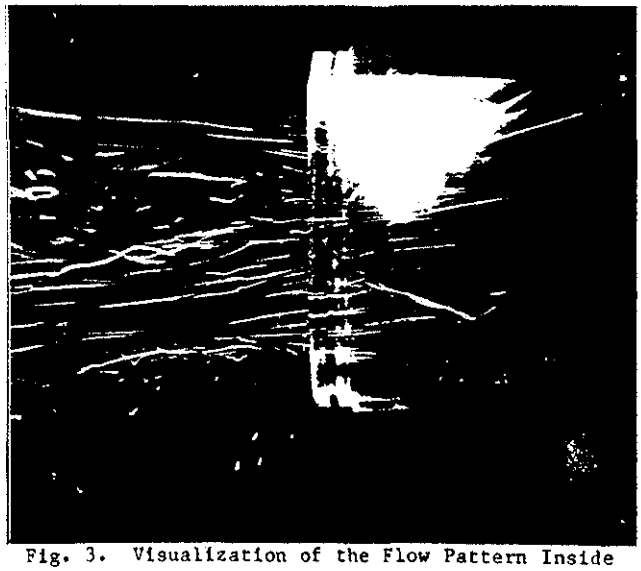

Fig. 3. Visualization of the Flow Pattern Inside the Tapered-Elliptic Nozzle (Water Tunnel Test).

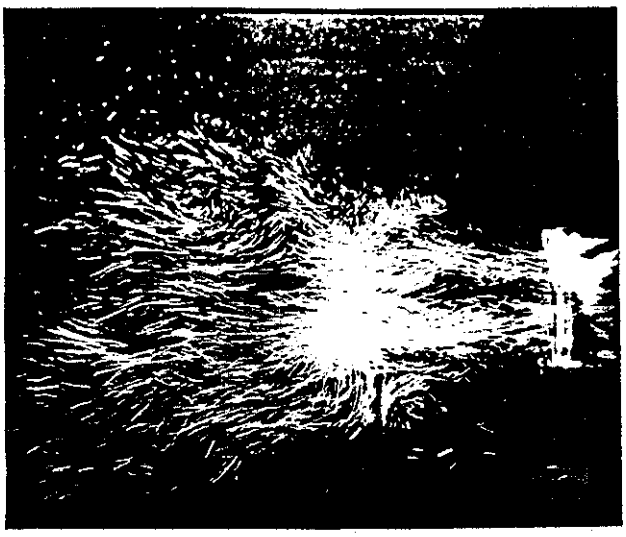

Fig. 4. Water Flow Visualization of Jet Issuing From the Tapered Elliptic Nozzle.

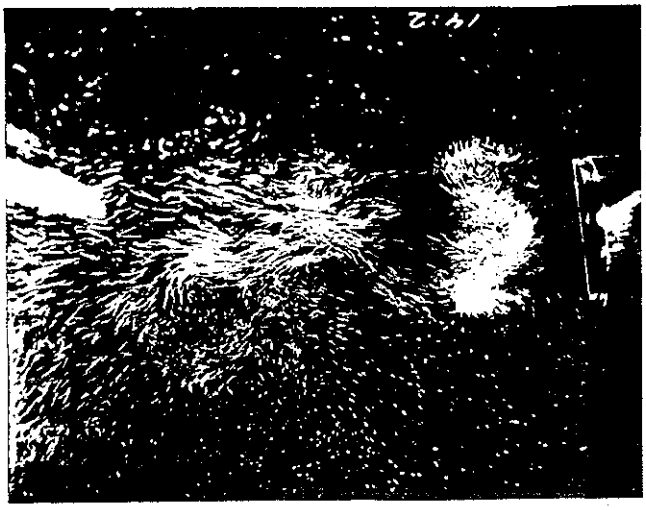

Fig. 5. Water Flow Visualization Demonstrating the Deformation of the Initial Vortices of the Tapered Elliptic Jet.

Flow Field

The mean and turbulent flow field of the tapered slot jet are compared to a circular jet in Figs. 6 and 7. The tapered slot jet has an accelerated mean velocity decay along the centerline, relative to a circular jet. At the same time, higher mean velocity extensions are developed in the jet's circumference, while the circular jet has a Gaussian shaped mean velocity in the radial direction, decreasing monotonically to zero at the jet's edge. The axial turbulence intensity contours, given in Fig. 7, show the rapid amplification of the turbulence level along the centerline of the tapered slot jet relative to the low rate of the circular jet. The turbulence growth along the two jets' shear layers are similar.

Two geometrical parameters of the nozzle design determine the jet characteristics: (1) the conical contraction angle (α), (2) and the aspect ratio (AR) of the outlet slot. In the following section the effect of these parameters is discussed.

Effect of Conical Contraction Angle

The change of the jet's spreading rate in the major and minor axes planes for different contraction angles is shown in Figs. 8 and 9, respectively. The spreading rate in the major axis plane increases with increasing the contraction angle. The jet with the smallest contraction angle ($\alpha = 7.5°$) has the same spreading rate as a circular jet. The rate is doubled for $\alpha = 30°$. In the minor axis plane (Fig. 9) the three tapered slot jets spread faster than the circular jet. The same trend of increased spread with larger α is also obtained here. The initial reduction of the jet width at the minor axis plane for $x/D_e \leq 2$ is due to the development of two secondary jets at the outer sections of the major axis.

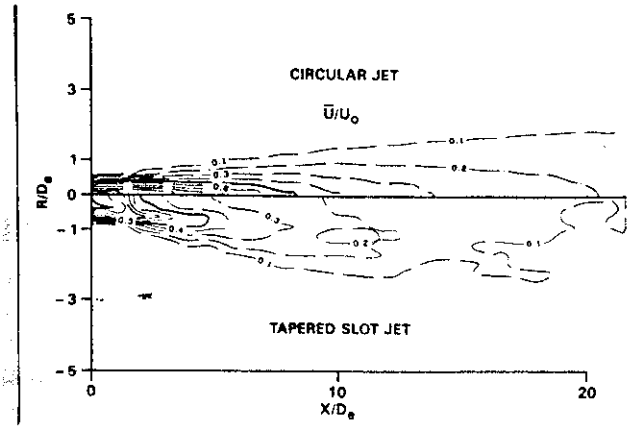

Fig. 6. Mean Axial Velocity Contours.

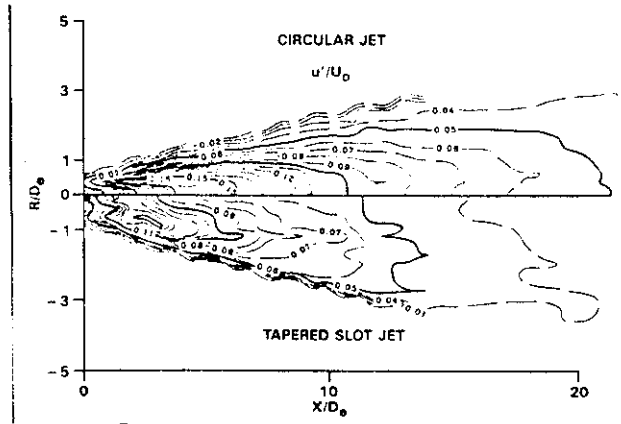

Fig. 7. Axial Turbulence Intensity Contours.

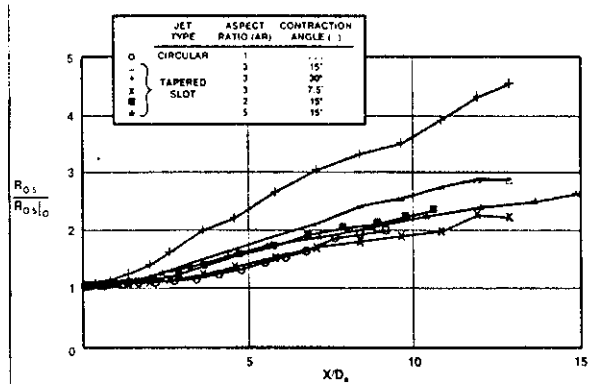

Fig. 8. Variation of Jet Spreading Rate With Contraction Angle and Aspect Ratio (Major Axis).

The variation of the spreading rate with the contraction angle affects the mean velocity decay along the jet centerline (Fig. 10). The highest decay rate correlates with the largest contraction angle, $\alpha = 30°$. The jet with the smallest angle, $\alpha = 7.5°$, is closest to the circular jet mean velocity change. The potential core length, as determined by the distance from the jet exit to the location where the velocity drops below the exit velocity, is shorter for all three tapered slot jets, $x = 1.3\ D_e$, relative to $3.5\ D_e$ for the circular jet. The increased velocity of the jets with $\alpha = 15°$ and $7.5°$ in this section, is due to the initial reduction of the jet's width at the minor axis side.

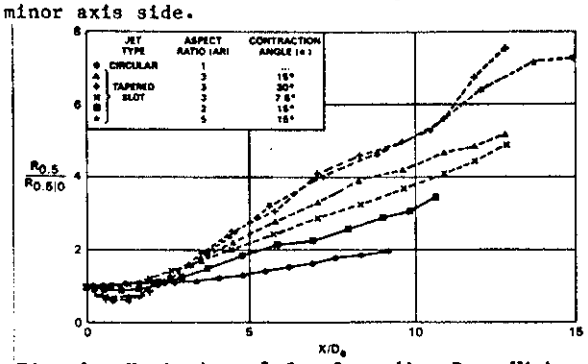

Fig. 9. Variation of Jet Spreading Rate With Contraction Angle and Aspect Ratio (Minor Axis).

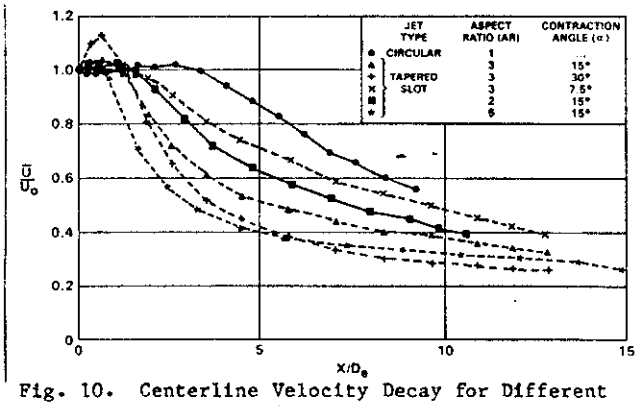

Fig. 10. Centerline Velocity Decay for Different Contraction Angles and Aspect Ratios.

One of the important features of the tapered slot jet is the higher augmentation of turbulence in the core region. The contraction angle affects this behavior too (Fig. 11). The maximum amplification is obtained for the steepest contraction angle, $\alpha = 30°$. The turbulence level is increased from an initial value of $u'/U_o < 0.05$ at the exit to $u'/U_o = 0.18$ at $x/D_e = 1.9$. The circular jet retains its initial level of 10% in this range and it reaches its highest level of 15% at $6.2\ D_e$. The jet with the intermediate angle, $\alpha = 15°$, reaches the same peak level as the circular jet at $1.9\ D_e$, and the jet with the smallest angle has a maximum turbulence level of less than that of the circular jet. Similar results were obtained for the other two components of the turbulent fluctuations, v' and w' (not shown here).

The turbulence intensity variation along the shear layer region at the major axis side (Fig. 12) is nearly equal for all three tapered slot jets and similar to the circular jet. The highest level of 18% is obtained at $2.6\ D_e$ for the tapered jets and at $1.5\ D_e$ for the circular jet.

At the minor axis shear layer (Fig. 13), the contraction angle changes the amplification rate. The jet with the largest angle, $\alpha = 30°$, has the same behavior as the circular jet, while the other two reach a level which is 15% lower than the circular.

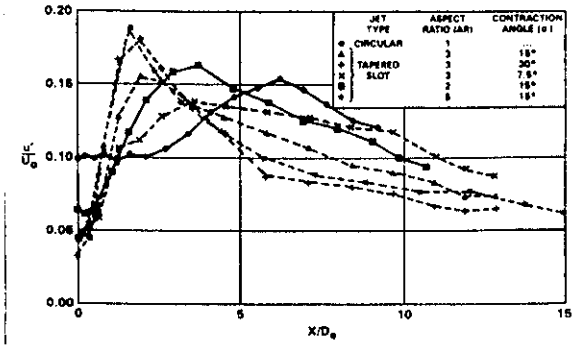

Fig. 11. Centerline Turbulence Intensity for Different Contraction Angles and Aspect Ratios.

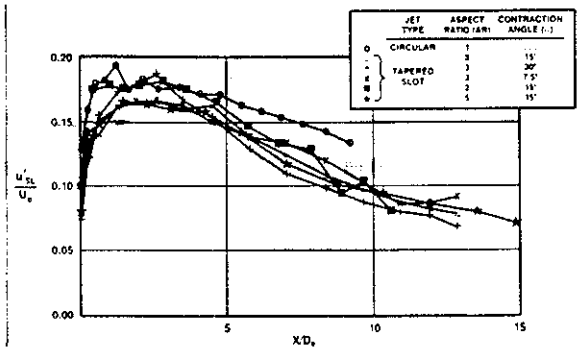

Fig. 12. Turbulence Intensity Variation Along the Jet Shear Layer for Different Contraction Angles and Aspect Ratios (Major Axis).

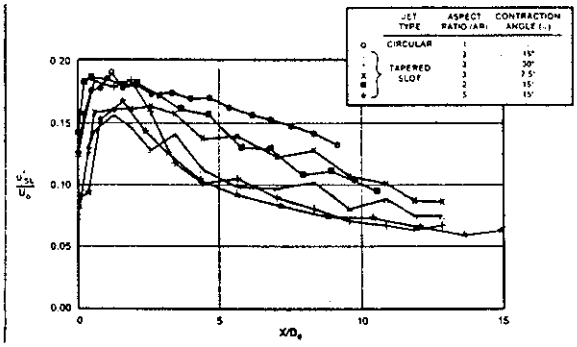

Fig. 13. Turbulence Intensity Variation Along the Jet Shear Layer for Different Contraction Angles and Aspect Ratios (Minor Axis).

Effect of the Aspect Ratio

Three tapered slot jets with aspect ratios of 2, 3, and 5 were tested and compared to a circular jet. All jets had a contraction angle of 15°. The jet spread at the major axis plane did not change with the aspect ratio and is slightly higher than the circular jet. The spread in the minor axis plane (Fig. 9) increases with the jet aspect ratio. The high aspect ratio jet (AR = 5) spreads almost twice as much as the AR = 2 jet. All tapered jets grow in the minor axis plane more than the circular jet.

The large aspect ratio jet also has a higher mean velocity decay rate along the centerline (Fig. 10). The potential core length is reduced from 1.6 D_e for the AR = 2 jet to 0.6 D_e for the AR = 5 jet.

The highest amplification of turbulence intensity along the centerline is for the AR = 5 jet (Fig. 11). Its rate is even higher than that of the jet with the steepest contraction angle ($\alpha = 30°$, AR = 3). The jet reaches a turbulence level of nearly 19% at a distance of 1.6 D_e from the nozzle. The amplification rate is reduced with decreasing aspect ratio, but is still higher than the circular jet even for the AR = 2 jet.

The amplification of the turbulent fluctuations in the shear layer are given in Figs. 12 and 13 for the major and minor axes of the tapered jet with different aspect ratios. The tapered jet with AR = 2 has the highest amplification rate at the major axis shear layer, which is also equal to that of the circular jet. The jets with asepct ratios of 2 and 3 have similar behavior and reach a maximum level which is lower than the circular jet. The same features are also observed at the minor axis plane (Fig. 13).

Reacting Flow

The tapered slot jet was tested in a premixed flame. Nozzle number 7 with an aspect ratio of 3 and contraction angle, $\alpha = 22°$, was used for the burner. Figure 14 shows the hydroxyl concentration in three axial cross sections of the flame. The hydroxyl radical is an intermediate and equilibrium product of the propane/air flame. Frames 1 and 2 are obtained by passing the laser sheet along the major axis, in the flame's axis direction. In frames 3 and 4, the beam is passing through the minor axis plane and in frames 5 and 6 the beam plane is parallel to the minor axis plane at a distance which is equal to one fourth of the major axis length, from the center. Frames 1, 3, and 5 differ from frames 2, 4, and 6 by the time delay between the laser pulse and the forcing signal, thus the flow is visualized in the two groups at different phases relative to the forcing. The OH concentration is higher in this flame relative to a circular burner. The reaction occurs both in the flame circumference and in the flame core. This is different than a circular premixed flame in which the combustion is concentrated in the outer edge of the air jet.[8] The reaction in the flame's core is sustained due to the improved small-scale mixing obtained by the axial vortices which are generated in this region. The structures in the jet's core are surrounded by shear layer vortices which can be seen in frames 1, 4, and 6.

The special character of the tapered slot jet core is visualized in a different cross sectional plane in Fig. 15. The laser sheet is "slicing" the flame perpendicular to the flame's axis. The two frames show an averaged picture of 100 single realizations (a) and an instantaneous picture of 18 nsec duration (b). The three-dimensional structures in the middle of the flame are cross-sections in the axial vortices emanating from the nozzle. The outer closed surface is a cross section of the flame's edge. These pictures show the increased OH concentration obtained by the additional turbulence generation gained by using this burner.

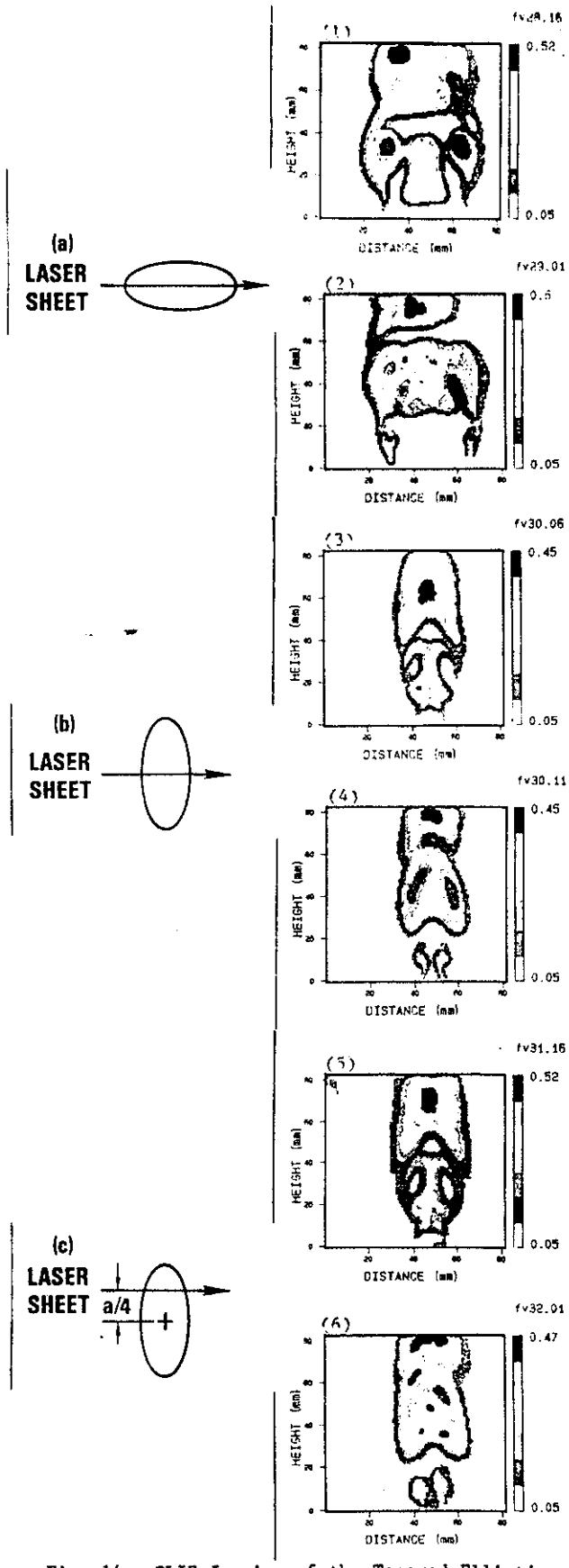

Fig. 14. PLIF Imaging of the Tapered Elliptic Flame in a Plane Parallel to the Flame's Axis.

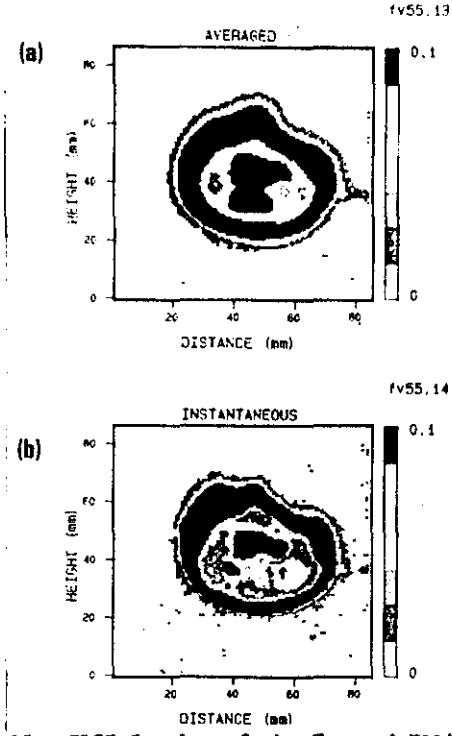

Fig. 15. PLIF Imaging of the Tapered Elliptic Flame in a Plane Perpendicular to the Flame's Axis.

Conclusions

A novel design aimed at reducing flow coherence and enhancing turbulence and small-scale mixing in the entire reactor is described in this paper. The regular elliptic jet, which was shown to have special mixing characteristics relative to a circular jet, is modified by adding a conical contraction leading to the elongated slot outlet. The combination of this contraction with the noncircular outlet have a substantial effect on the outgoing jet flow. The elliptic jet is characterized by its axes switching in the downstream direction. The initial elliptic jet changes shape very rapidly due to the high spreading rate at the minor axis section. The initial minor axis becomes the jet's new major axis and vice versa at the other side. The flow of the tapered slot jet is very different. The spreading rate on the side of the minor axis is reduced, while the growth along the major axis is increased considerably. At the same time, the flow accelerates at the major axis plane resulting in the formation of a double-jet flow.

The centerline turbulence level of the tapered slot jet is up to five times higher than in the circular jet. In addition to the highly turbulent flow in the vicinity of the jet center, the high turbulence activity in the jet's shear layer, i.e., at the jet periphery, is maintained.

These flow characteristics are related to several different causes. The introduction of the nonsymmetric azimuthal disturbances into the initial shear layer of the jet results in a very complicated flow field. The flow is unstable to numerous azimuthal modes leading to an eventual breakdown of the flow into small scale turbulence, thus enhancing the mixing at the fine scale. The double shear layer generated inside the flow due to the two jets which are formed in the major axis plane contribute to the additional production of turbulence in the shear layer as well as in the core region. The third cause is probably the dominant one and is related to the inside structure of the nozzle. The intersection between the conical contraction and the slot introduces axial rotation into the outgoing flow. The jet flow injected from the nozzle has, in addition to the spanwise vorticity, an axial component. The angular momentum of the axial vortices is determined by the transition surface between the conical contraction and the slot outlet. It is shown that by either increasing the conical contraction angle or by increasing the slot aspect ratio, the intensity of the axial vortices is increased and the ensuing effect on the mean flow field is higher. These two geometrical changes have similar effect on the above mentioned transition surface. The two parameters can be used to control and vary the mean and turbulent flow of the tapered slot jet as desired. The spreading rate of the jet at the two axis planes can be altered as well as the turbulence level in the jet's core and shear layers.

The structure of the tapered slot burner flame is influenced by this special flow field. Intense combustion is observed both in the shear layer and core regions as expected from the cold flow results. Because of the enhanced small-scale mixing of this jet the flame efficiently burns the fuel without soot formation even at high equivalence ratios.

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